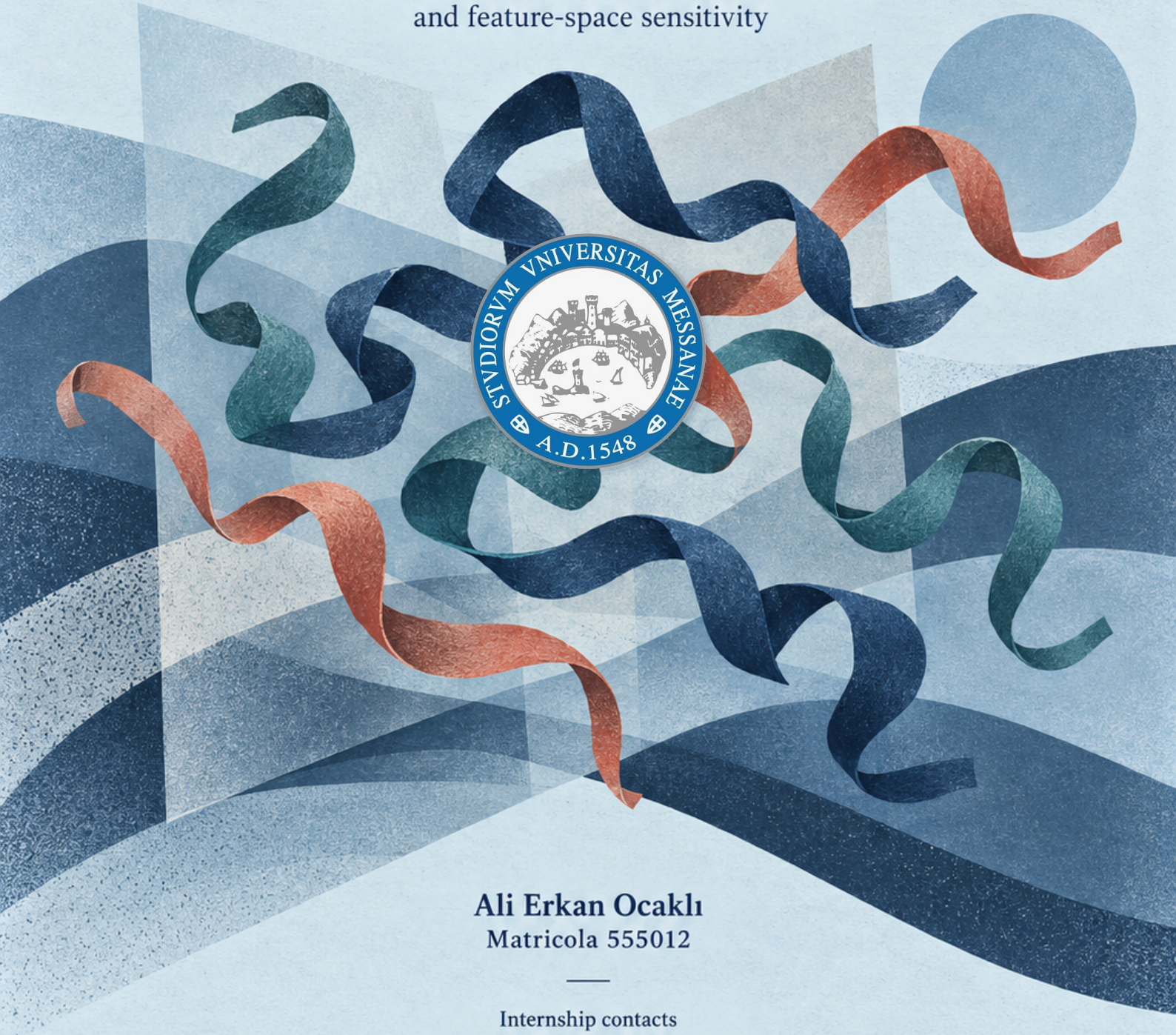
UNIVERSITY OF MESSINA
Data Analysis L-31

CURRICULAR INTERNSHIP REPORT

RobustCluster-Aβ

Reproducible analysis of amyloid-β40 conformational descriptors

Scientific code audit, sampling diagnostics and feature-space sensitivity



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Abstract

Molecular simulation data require both statistical analysis and an understanding of how the observations were generated. This curricular internship used amyloid-β40 descriptors to connect practical work with GROMACS, VMD and Python to a reproducible analysis of conformational clustering. The report develops the scientific background, examines the supplied implementation of the workflow of Cadenelli and colleagues, and presents a separate experiment on the choice of clustering features.

The available table contains 300,003 observations from three replicas over the source-labelled 1–3 microsecond interval. The audit identified problems in input handling, residual-cluster treatment, free-energy interpolation and weighted summaries. A modular implementation addresses these operations through explicit validation, numerical tests and recorded run settings. The recovered free-energy surface contains the complete grid and covers all converted descriptor queries. Its energy units and construction procedure remain unverified, so equilibrium clustering and population estimates are not reported.

The completed descriptor analysis found a pooled correlation of 0.848 between radius of gyration and solvent-accessible surface area (SASA). The first two standardized principal components account for 88.1% of variance. To examine whether the correlation justified removing SASA, Gaussian mixtures were fitted to four and three descriptors using one shared, replica-balanced fitting sample. Five component counts and five initialization seeds produced fifty converged fits. The median agreement between feature sets was 0.864 at three components and 0.457 at six, showing that the effect of feature removal depends on the fitted model.

All twelve replica–descriptor correlation estimates failed at least one reliability check. The results therefore remain descriptive comparisons in a biased, temporally dependent sample; they do not establish physical metastable states or unbiased kinetic rates. The accompanying code, figures and source records make the completed calculations inspectable. The learning account also identifies VMD MCP and GROMACS MCP as personal software projects, without attributing the supplied simulations to them.

Keywords: amyloid-β40; intrinsically disordered peptides; well-tempered metadynamics; density peaks; statistical reweighting; reproducibility.

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Abbreviations and evidence categories

Term	Meaning
ACF	Autocorrelation function
ARI	Adjusted Rand index
CV	Collective variable
GMM	Gaussian mixture model
DPC / DPA	Density Peak Clustering / Density Peaks Advanced
ESS	Effective sample size; estimator and assumptions stated explicitly
FES	Free-energy surface
IDP	Intrinsically disordered protein or peptide
MD	Molecular dynamics
PCA	Principal component analysis
SASA	Solvent-accessible surface area
SEM	Standard error of the mean
WT-MetaD	Well-tempered metadynamics

The report separates the published method, its implementation, corrections to the supplied code, additional robustness experiments, and future work. A software test checks how the code behaves on a specified example; it does not establish a biological result. An analysis marked BLOCKED lacks required inputs or information about their origin. Learning topics in the internship programme are also distinguished from work that has a recorded output.

1 Introduction and internship objectives

1.1 From data analysis to molecular ensembles

Molecular descriptors form a familiar data-analysis problem: each row contains measurements, and clustering groups similar rows. Their interpretation requires care. Consecutive observations may be correlated, simulation bias changes how often configurations appear, and descriptors can hide structural differences. These properties matter when choosing how to group the data.

This was the setting of the Data Analysis L-31 internship at the University of Messina, with Andrea Ciccolo and Valeria Conti Nibali as internship contacts. The programme introduced protein structure, intrinsic disorder, force fields, molecular dynamics and clustering. Practical work included GROMACS, VMD and Python. The final assignment combined analysis of supplied simulation data with an independent contribution using methods from the degree programme.

An intrinsically disordered peptide is described through many configurations rather than a single fixed structure. Disorder does not imply an absence of structural organization [1]. Clustering summarizes this diversity, but similarity in measured descriptors alone does not establish a physically distinct conformational family.

1.2 The Aβ40 case study

Cadenelli and colleagues studied monomeric Aβ40 and Aβ42 ensembles using well-tempered metadynamics, density-peak clustering, consensus between local clusters and statistical reweighting [2]. Metadynamics encourages exploration by adding a bias along selected collective variables; the well-tempered formulation controls its development [3]. Frequencies observed under this bias are not automatically equilibrium probabilities.

The supplied materials concern Aβ40: a clustering script, a free-energy surface and 300,003 descriptor observations in three replica segments. The four baseline features describe secondary-structure content, radius of gyration and solvent-accessible surface area (SASA). These measurements make configurations manageable as numerical data; they are not atomic coordinates.

The report audits the supplied workflow against the paper and separately examines feature choice. The alternative experiment does not reproduce the published results.

1.3 Questions and scope of the report

1. **Can the analysis be reconstructed?** Make frame identity, scaling, FES interpolation and weighted summaries explicit and testable.
2. **What do the descriptors reveal?** Examine redundancy and temporal dependence through correlation, PCA and autocorrelation diagnostics.
3. **Does removing SASA preserve the fitted groups?** Compare Gaussian mixtures across feature sets, component counts and initialization seeds on shared observations. Correlation with radius of gyration motivates this question but does not answer it; components are not identified as physical states.

The FES passes grid checks, but its energy units and construction remain unverified; equilibrium populations are not reported. The following chapters connect theory and methods to recorded results and learning outcomes. Code, tables and run records make the calculations inspectable [4, 5].

2 Scientific and theoretical background

2.1 Amyloid-β40 and intrinsically disordered proteins

The object being clustered is a set of peptide conformations, not a set of different proteins. Aβ40 belongs to the intrinsically disordered systems for which one structure cannot adequately describe the ensemble [1, 2]. An ensemble includes both the possible conformations and their statistical weights. This distinction matters when comparing a compact configuration with an extended one: observing both does not tell us how probable either is at equilibrium.

Experiments also often report averages over conformations. Combining simulation with NMR or SAXS constraints can help determine an ensemble, provided that the calculated and measured observables are compatible [6].

2.2 Molecular dynamics and conformational ensembles

Molecular dynamics follows atomic positions by numerically integrating equations of motion. A force field supplies the potential energy and hence the forces; the solvent model and thermodynamic conditions also affect the result. The primary study used GROMACS, the CHARMM36m force field and PLUMED [2, 7–9]. Here, the supplied descriptors are the starting point for analysis.

At canonical equilibrium, a configuration with potential energy $U(q)$ contributes according to the Boltzmann factor $\exp[-\beta_T U(q)]$. Coordinates are denoted by q. The inverse temperature factor is $\beta_T = 1/(k_B T)$ for energy per molecule and $1/(RT)$ for molar energy. The energy convention must therefore be known before weights can be calculated.

2.3 Sampling limitations

A trajectory with many saved frames may contain far fewer independent observations. If successive conformations change slowly, saving them more frequently mainly adds similar rows. Correlation analysis estimates this dependence under stated assumptions; it cannot recover structures that were never visited. Equilibration and sampling quality consequently need more evidence than trajectory length alone [10, 11].

This problem becomes especially important when estimating uncertainty. Treating dependent frames as independent can make an estimate appear more precise than the available sampling supports.

2.4 Well-tempered metadynamics

Metadynamics encourages exploration by adding a potential along a small set of collective variables (CVs), numerical coordinates chosen to describe relevant changes. The bias depends on where the system has already been. In well-tempered metadynamics, its growth slows as bias accumulates. The bias factor is $\gamma = (T + \Delta T)/T > 1$; in the long-time regime the bias approaches a scaled negative free energy, up to a constant [3].

The method changes both forces and sampled frequencies. Statistical reweighting is therefore needed for equilibrium populations. The recorded trajectory time cannot directly supply unbiased relaxation times or transition rates.

2.5 Gaussian hills and bias construction

The accumulated potential can be written as a sum of local Gaussian hills:

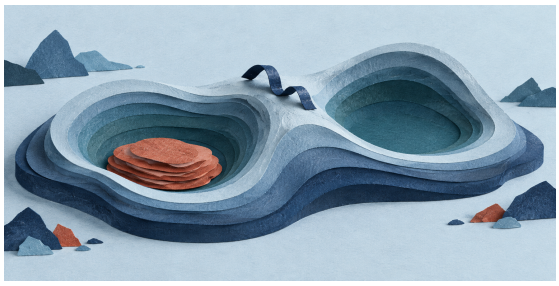
$$V(s,t) = \sum_{t_k<t} h_k \exp\left[-\frac{1}{2}\sum_a \frac{(s_a - s_a(t_k))^2}{\sigma_a^2}\right]. \tag{2.1}$$

In eq. (2.1), s is the CV vector, t_k a deposition time, h_k the hill height and σ_a its width along coordinate a. Well-tempered heights follow $h_k = h_0 \exp[-V(s(t_k), t_k)/(k_B \Delta T)]$ in molecular energy units; molar units require R instead of k_B.

PLUMED reports the instantaneous bias as `metad.bias`. Its offset-corrected output is `metad.rbias` = `metad.bias` − `metad.rct`, where the last term is $c(t)$ [12]. Understanding these distinct quantities is necessary before interpreting an exported bias column.



(a) A conformational ensemble



(b) Accumulating sampling bias

Figure 1: Conceptual editorial illustrations of two ideas used in the analysis. Flexible ribbons suggest conformational diversity; layered deposits suggest the accumulated bias in eq. (2.1). These generated images contain no atomic coordinates, measured free energies or trajectory results. Their provenance is recorded with the report sources.

2.6 Free-energy surfaces

For a CV probability density $p(s)$, free energy is $F(s) = -\beta_T^{-1} \log p(s) + C$. The constant C fixes an arbitrary reference. Integrating over coordinates omitted from the CVs can conceal structural differences: several atomic arrangements can occupy the same location on a two-dimensional surface.

PLUMED's `sum_hills` reconstructs surfaces from hill records and distinguishes free-energy output from negative-bias output [13]. Confusing them can apply the well-tempered scaling factor twice. A valid grid and a documented physical interpretation are therefore separate requirements.

2.7 Statistical reweighting

Under a quasi-static bias, multiplying sampled contributions by weights proportional to $\exp(\beta_T V)$ corrects their relative probabilities. Combining this expression with the well-tempered relation gives the final-bias approximation used in the primary study:

$$\ell_i \equiv \log w_i = -\frac{1 - 1/\gamma}{RT} F(\alpha_i, \beta_i). \tag{2.2}$$

Equation 2.2 assumes molar energy units and an interval for which the bias can be treated as sufficiently stable. Adding a constant to F shifts all log weights equally; normalization removes that shift.

Other estimators account explicitly for the changing bias normalization [14, 15]. With verified PLUMED offset-corrected output, the optional comparison uses $\ell_i = \mathrm{rbias}_i/(RT)$ [16]. This is a separate estimator, not another name for final-bias weighting.

2.8 Structural descriptors

The four baseline descriptors summarize secondary structure, compactness and solvent exposure. The supplied n_α and n_β are scaled continuous CVs that approximate residue counts. They are not exact integer assignments from DSSP, which uses geometrical and hydrogen-bond information [17]. Radius of gyration, R_g, measures spatial spread, while SASA measures accessible surface area.

These descriptors also have different units and definitions. Radius of gyration is a length, whereas SASA is an area. In a Euclidean distance between raw descriptor rows, changing a length unit would change that coordinate's contribution without changing the molecule. Normalization makes the comparison explicit, but it does not prove that all four descriptors contain equally useful structural information. The choice of representation is therefore part of the method, not just a plotting preference.

Similar R_g and SASA values need not imply identical residue contacts. Their possible redundancy is a question to test, not an automatic consequence of correlation. Apolar/polar SASA and hydrogen-bond descriptors can provide additional characterization after clustering, once their definitions are established.

2.9 Density-based conformational clustering

A density peak is a point surrounded by many neighbours yet separated from points of higher density. The published workflow repeatedly searches for such centres in random subsets and retains clusters satisfying size and silhouette criteria [2]. The silhouette compares proximity to one's own cluster with proximity to other clusters [18].

The silhouette of a frame compares its average distance to its own cluster, a, with the smallest average distance to another cluster, b: $(b-a)/\max(a,b)$. A positive value favours its assigned cluster in the chosen geometry; a value near zero indicates a boundary region. A mean threshold filters the local groups, but it need not reject every ambiguous frame. This explains why the final assignment remains a separate operation.

These geometric criteria make clustering inspectable, but do not establish metastable physical states. Consensus groups local clusters into broader families; final assignment and statistical weighting answer the separate questions of which frames belong to each family and how much each contributes.

3 Data and computational methodology

3.1 Data provenance and scope

The analysis begins with a supplied text table, not with atomic coordinates. Each row contains time, an offset-corrected bias field and eight structural measurements. The file has 300 003 rows and ten columns. The original header is retained in the inventory, including its spelling of the peptide–solvent hydrogen-bond field. The script, descriptor table and recovered FES have recorded SHA256 hashes, which allow later copies to be checked against the materials inspected here.

The published method provides the scientific reference [2]. The supplied script provides additional implementation details, some of which differ from that description. The reproduction therefore has two sources of information: the paper for the intended calculation, and the code for choices that need to be checked. The independent GMM experiment has its own configuration and does not change the reproduction baseline.

3.2 Replica segmentation and analysis interval

A decrease in the time column marks the beginning of a new replica. Validation requires exactly three segments, strictly increasing times within each segment, finite entries and complete coverage of the selected interval. It rejects malformed rows rather than discarding them. Each accepted row retains its original file line, source-row index, replica ID and frame index within that replica. These identities are needed when two runs assign labels in different orders.

The analysis includes both endpoints from 1 000 000 to 3 000 000, giving 100 001 frames per replica at spacing 20. Interpreting these values as picoseconds follows the supplied script and published protocol, so the interval is recorded as 1–3 µs with that unit assumption. The first microsecond is not in the table. Its exclusion can be reproduced, but its equilibration cannot be reassessed from these data.

3.3 Descriptor definitions and normalization

The baseline features are n_α, n_β, R_g and SASA. The first two are continuous secondary-structure descriptors. The original script labels R_g in nm and SASA in nm^2; their upstream calculation settings remain unavailable. The additional SASA and hydrogen-bond columns are intended for later cluster characterization.

Clustering uses pooled min–max scaling. For descriptor x_f, the transformed value is $z_f = (x_f - a_f)/(b_f - a_f)$, where a_f and b_f are the minimum and maximum across the complete analysis interval. The saved transformation gives every replica and discovery group the same coordinate system. A constant feature is rejected because the denominator would be zero. The GMM experiment instead standardizes its fitting sample, as described in Chapter 5.

3.4 STEP 1: local Density Peak Clustering

Figure 2 summarizes the implemented reproduction. Frames are divided into random, nonoverlapping groups of approximately 30 000. A final group of at most ten frames is redistributed over the preceding groups. Local NumPy generators derive their streams from a recorded master seed

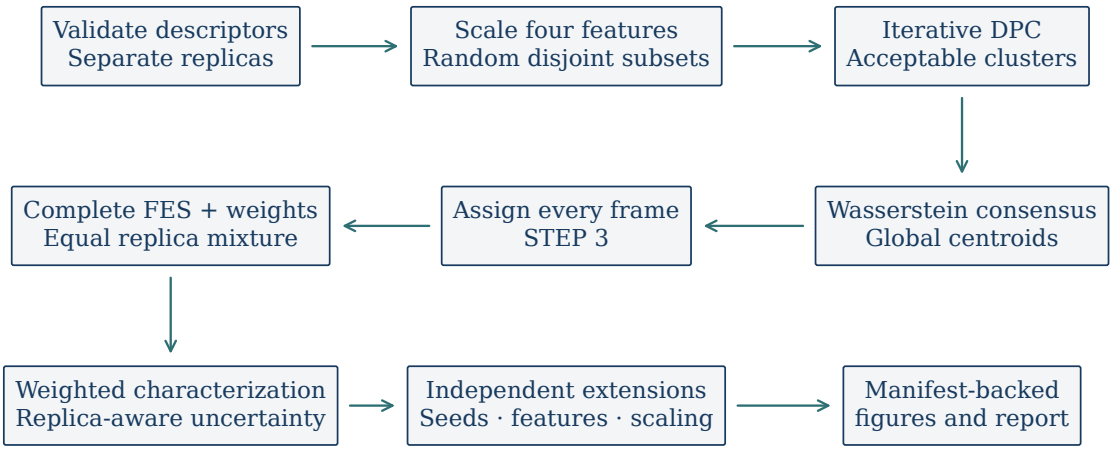


Figure 2: Implemented reproduction workflow. Local accepted clusters define global families; every analysis frame is subsequently assigned a label. The independent extensions have separate configurations. The diagram describes the software design: complete empirical reweighting has not been performed.

[19]. Their draws are reproducible, although they do not reproduce the legacy global-RNG sequence exactly.

Within a group, local density ρ_i counts neighbours inside d_c, including the point itself. Points are ordered by decreasing density, with local frame position breaking ties. Separation δ_i is the distance to the nearest earlier point in that order; the first point uses the maximum pairwise distance. The ordering rule is needed because equal-density points are not strictly denser neighbours. These quantities enter the following iterative protocol.

1. **Generate candidates.** Start discovery at $d_c = 1.3$, rank candidate centres by $\rho_i\delta_i$, and try 10 down to 2 centres.
2. **Check each cluster.** Require at least 300 members and a mean silhouette of at least 0.3 for acceptance.
3. **Choose the candidate.** Keep the one with the most acceptable clusters. A tie retains the earlier, higher-centre-count candidate.
4. **Continue on the remainder.** Remove accepted members and multiply d_c by 0.9 before the next iteration.
5. **Stop discovery.** Stop below 300 remaining points or after five consecutive unsuccessful iterations. Residual frames receive their final labels only in STEP 3.

Pairwise distances are stored as float32 values in a matrix constructed in blocks. Density counts do not allocate a second floating-point kernel matrix. Silhouette calculations reuse the distances where the label count is valid, between two and $N - 1$ [20]. Storage still grows as N^2; these changes reduce extra memory rather than remove the quadratic cost.

3.5 STEP 2: Wasserstein consensus clustering

Consensus includes accepted local clusters only. For clusters A and B, the supplied-code distance is

$$D(A,B) = \left[\sum_{f=1}^{4} W_1(A_f, B_f)^2\right]^{1/2}. \tag{3.1}$$

In eq. (3.1), each W_1 compares the observed distributions of one descriptor [21]. The combined distance is not joint four-dimensional optimal transport: two clusters may have similar individual distributions but different relationships between descriptors.

The baseline retains the script's raw descriptor scales. Because this combines different units and ranges, normalized Wasserstein distances are reserved for a separate sensitivity experiment. Average-linkage clustering is cut using the source's cubic-spline elbow heuristic: smoothing parameter 5 and maximum second derivative within the first 90% of merges. Fewer than five accepted clusters or a flat merge-height curve causes an explicit failure.

3.6 STEP 3: assigning every frame

A global centroid is the mean normalized descriptor vector of its accepted local-cluster members. Every analysis frame is assigned to its nearest centroid, including residuals and frames outside reduced discovery groups. Ties use deterministic label order. Centroids are not updated after assignment, since an iterative k-means refinement would change the specified method. Labels are saved with frame identities for later statistics and comparisons.

3.7 FES interpolation and statistical weights

The FES reader checks header dimensions, axis bounds, nonperiodic status, coordinate spacing, duplicate pairs and finite values, including derivative columns. All expected grid points must be present. Row order is unrestricted, but partial-download files are rejected. Descriptor queries use $\alpha = n_\alpha/1.1$ and $\beta = n_\beta/2.5$; these are physical CV conversions, not the clustering normalization.

Bilinear interpolation uses `RegularGridInterpolator` with its linear method and strict bounds checking [22, 23]. Points outside the grid are errors, not candidates for clipping. An analytic surface with a mixed $\alpha\beta$ term tests the interpolation between grid nodes as well as at the boundaries.

The final-bias expression in eq. (2.2) is evaluated in log space. For replica r, weights are normalized before pooling:

$$\widetilde{w}_{ri} = \exp\left[\ell_{ri} - \mathrm{logsumexp}_j(\ell_{rj})\right], \qquad W_{ri} = \widetilde{w}_{ri}/3. \tag{3.2}$$

Equation 3.2 gives each replica one third of the total weight. Raw exponentials are saved for inspection with overflow or underflow flags; normalized calculations continue to use log weights. The intended frame table records identities, time, CVs, interpolated energy, raw and normalized weights, and final labels. It has not been produced for the real data because the FES energy units and construction procedure remain unverified.

3.8 Cluster summaries and reproducibility

A family's relevance is the sum of its pooled member weights. Conditional means and distribution standard deviations use the same weights within that family. Distribution width describes variation among members; replica SEM describes uncertainty in average relevance across replicas. The optional block bootstrap resamples consecutive observations separately within each replica while holding labels and the FES fixed. Its intervals would not include uncertainty from reconstructing the FES or refitting clusters.

NumPy, SciPy, pandas, Matplotlib and scikit-learn support the modular implementation [23–25]. Commands and the Snakemake workflow use the same functions [5, 26]. Before execution, a manifest records configuration, hashes, seeds and software versions. Runtime and sampled process-tree memory are logged, and resource limits are enforced. Successful, failed and blocked stages remain distinguishable. This record is what connects a reported figure to a particular calculation.

4 Reproduction and scientific code audit

4.1 Original implementation

The audit began by separating three questions: what the paper describes, what the supplied code attempts, and what execution records demonstrate. The original combines input reading, replica separation, clustering, weighting and plotting in one script. A bare Italian sentence at line 437 causes a syntax error before execution. The file is preserved unchanged, so this finding and the later methodological comparisons remain checkable.

The script is evidence of implementation intent, not a completed run. No original execution manifest or complete output set was located. The method map consequently records paper statements and source behaviour separately.

4.2 Identified methodological issues

The corrections in table 1 concern which observations enter the analysis and how clusters are interpreted. They are separated from changes made solely to organize the software.

Supplied-script behaviour	Correction and consequence
Takes the first three time-reset segments and skips some malformed rows.	Checks every row and requires exactly three segments, making incomplete or invalid input an explicit failure.
Sends rejected frames into consensus under a generic label.	Uses accepted memberships to discover global centres. Rejected frames receive final labels without being treated as a discovered family.
Permits missing FES cells; nearest-grid lookup can return a boundary value for an out-of-range query.	Requires a complete grid and bounded bilinear interpolation; queries outside the grid fail.
Weights relevance but uses unweighted descriptor means and standard deviations.	Weights cluster summaries consistently and normalizes each replica before pooling. Distribution width, replica SEM and bootstrap intervals remain distinct.

Table 1: Methodological corrections in the modular implementation. The original script remains unchanged.

Random grouping required a separate correction. The original would exhaust its fixed list of 17 seeds if sufficiently many groups were requested. Local random-number streams remove that limit, while changing the exact subsets drawn by the legacy code. For the FES, the partial download remains rejected and the complete replacement passed structural validation. Neither that validation nor the interpolation correction establishes its energy unit or reconstruction procedure.

4.3 Scientific V2: unavailable intermediate evidence

Neither `CLUSTERING40_scientific_v2.py` nor its changelog was located. The present corrections belong to the modular implementation; they are not attributed to an unseen intermediate version. If recovered, V2 should be preserved, hashed and compared directly with both available versions.

4.4 Verification and numerical tests

The selected release record reports 36 passing tests. Two small examples show how a check can identify a numerical error without relying on the output of the full pipeline.

Interpolation with a known answer. An analytic bilinear surface includes a mixed-coordinate term, so a query between nodes tests both axes together. Reordering its rows must preserve the interpolated answer. Deleting or duplicating a grid point, or querying outside the bounds, must instead raise an error. The same fixture therefore checks both a valid calculation and the rejection of inputs for which that calculation would be inappropriate.

Normalization with an invariant. An arbitrary additive FES constant must disappear after weight normalization. A second check gives two replicas unequal numbers of frames and requires each to contribute half of the pooled mass. This catches pooling by row count, which would look correct on three equal-length replicas. Other weight tests check the formula's sign and weighted moments.

The remaining checks cover parsing, scaling, residual exclusion and final assignment. Renaming cluster labels must leave partition agreement unchanged; changing evaluation extremes must not alter a scaler fitted on fixed training rows. These checks isolate failures that could otherwise be mistaken for a change in the clustering method.

A synthetic end-to-end fixture exercises discovery, consensus, weighting, summaries and figure generation. Synthetic GMM cases check repeatability and missing-family handling. An autoregressive fixture probes the correlation estimator; an archive test prevents an unrelated enclosing Git repository from supplying false provenance. The fixtures establish software behaviour on known inputs. They are not additional Aβ40 experiments or evidence that inferred groups have biological meaning.

4.5 Remaining methodological ambiguities

The paper does not fully specify STEP-2 scaling, the replica-averaged FES construction or the conventions behind every reported width and uncertainty. Density ties and elbow selection are also clearer in the supplied code. Source-informed choices are therefore identified explicitly. Raw Wasserstein scaling remains the reproduction setting, with normalization reserved for a separate sensitivity experiment. Resolving these choices is necessary before differences from published populations can be interpreted scientifically.

5 RobustCluster-Aβ: robustness extension

5.1 Temporal correlation and effective sampling

The sampling diagnostic asks how much information repeated observations add. For each replica and descriptor, the mean is removed before FFT-based autocorrelation with common $1/N$ covariance normalization. Adjacent positive-lag terms are paired, truncated at the first nonpositive pair sum and adjusted by cumulative minima. The retained contributions define

$$g = \max\left(1, 1 + 2 \sum_{k \in \mathscr{K}} \hat{\rho}(k)\right), \qquad N_{\mathrm{eff,corr}} = N/g, \qquad \tau_{\mathrm{int}} = g\Delta t/2. \tag{5.1}$$

Here $\mathscr{K}$ identifies the lag contributions retained after this processing; eq. (5.1) includes the zero-lag half term in the integrated time. Reliability checks require a stopping lag, $N \geq 50g$, and a half-window mean difference no greater than 0.25 full-series standard deviations. These are screening heuristics, not a formal stationarity test [10, 11].

Unequal statistical weights require a separate diagnostic:

$$N_{\mathrm{eff,Kish}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{5.2}$$

Equation 5.2 measures weight concentration without correcting temporal dependence. It is not automatically combined with correlation ESS. No empirical Kish ESS is available without validated weights.

5.2 Cluster stability across seeds

ARI compares partitions independently of their label names [27]. Matching families by maximum overlap then supports Jaccard and centroid comparisons while retaining unmatched families. Seed variation measures algorithmic sensitivity, not uncertainty between independent replicas. The planned DPC study would repeat discovery, consensus and assignment together; its two-seed smoke, five-seed pilot and 20–50-seed full studies remain separate from the completed GMM experiment below.

5.3 Feature redundancy and PCA

The completed diagnostics combine Pearson correlation with PCA of standardized descriptors. Explained variance concerns the observed feature geometry, not equilibrium relevance. The alternative PCA clustering configuration retains all components, so rotation alone preserves Euclidean distances. Whitening or changing the initial scaling can change those distances; discarding information is not the explanation when all components remain.

5.4 Feature ablation: the completed GMM experiment

The independent experiment tests whether removing SASA changes assignments. It compares four descriptors with n_α, n_β and R_g using Gaussian mixtures:

$$p(\mathbf{x}) = \sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \Sigma_j), \qquad \sum_j \pi_j = 1. \tag{5.3}$$

In eq. (5.3), components approximate the sampled descriptor distribution; their mixture coefficients are not equilibrium weights. The scikit-learn implementation uses full covariance matrices [25].

The protocol was committed before execution. Its four operations separate the choice of fitting observations from the comparisons made afterwards:

1. **Fix the sample.** Select 1,000 frames per replica with seed 20261007. Use the same 3,000 observations for every fit.
2. **Fit the transformation.** Estimate means and population standard deviations on those rows only. Share the transformation of retained descriptors between the four- and three-feature representations.
3. **Fit the models.** For each $K = 2, \ldots, 6$, use seeds 42–46: 50 fits overall. Each fit uses one initialization, at most 500 iterations, tolerance 10^{-3} and covariance regularization 10^{-6}.
4. **Compare assignments.** Label all 300,003 frames, including fitting rows. Use five paired-seed ARIs to compare representations and ten seed pairs to compare initializations within each representation.

Saved outputs include label matching, Jaccard overlaps, raw replica fractions and centroid shifts in the shared standardized three-feature space. The evaluation is descriptive, not held-out prediction. Keeping the sample fixed excludes sample-selection uncertainty; observed ranges are not confidence intervals. BIC is compared only within a feature space.

5.5 Wasserstein scaling sensitivity

The planned comparison changes descriptor scaling while keeping the same four marginal distances. This tests whether a variable's numerical range dominates consensus. Raw scaling remains the baseline; normalization does not resolve what the authors intended. Joint or sliced distances, energy distance and MMD remain optional future alternatives.

5.6 Reweighting sensitivity

A future comparison would keep labels fixed while contrasting final-bias weights with verified `metad.rbias` weights [14–16]. Changes in descriptor distributions or family relevance would then reflect weighting. Unresolved units and generating settings currently block this calculation.

5.7 Alternative density-based clustering

DPA offers a different density-based benchmark [28], but no fit has been performed. Weighted ShapeGMM instead operates on structural coordinates [29]; scalar descriptor rows cannot replace the missing trajectory, atom selections and frame mapping needed for that comparison.

5.8 Optional external validation

An experimentally constrained Aβ40 ensemble could test agreement in compactness or secondary structure [6]. Sequence, temperature, atom selections, weights and observable definitions must first agree. No external ensemble has been processed, so this remains a proposed comparison rather than a validation result.

6 Results

6.1 What the available data support

The calculations reported here use the supplied Aβ40 descriptors. Tests with generated data are discussed with the software validation and are not included as molecular observations. Table 2 distinguishes the completed analyses from the unresolved reproduction steps.

Analysis	Status and scope
Descriptor diagnostics	Complete; unweighted observations from three replicas
FES grid and query checks	Complete; energy units and construction unresolved
GMM feature comparison	Complete; 50 fits on a fixed fitting sample
Paper DPC and reweighting	Empirical reproduction not completed
Structural representatives	Unavailable without coordinates and frame mapping

Table 2: Scope of the recorded calculations. Numerical validation of an input or implementation does not by itself establish an equilibrium result.

Input validation found three segments of 100 001 rows. Times increase at the source-labelled spacing of 20 ps, and the data satisfy the column, finite-value and interval checks. The descriptor calculation, run `20260921T011959-583b4421`, took 2.28 s with a sampled peak worker-process memory of 418.2 MiB. These measurements describe the local diagnostic run, not the expected cost of a full DPC analysis.

The FES check resolved the earlier incomplete-download problem. The recovered file contains 2,801 by 1,301 grid points, giving 3,644,101 rows with five finite columns. Every expected coordinate pair is present and none is duplicated. The nonperiodic axes cover 0–28 and 0–13 at spacing 0.01. All 300,003 converted queries are inside the grid and yield finite bilinear interpolations. The file is numerically usable, but its energy column declares no unit. That unresolved information prevents interpretation of calculated equilibrium weights.

6.2 Shared variation in the descriptors

The pooled Rg–SASA correlation is 0.848, compared with -0.456 between the two secondary-structure descriptors. Thus the strongest association among these pairs concerns compaction and solvent exposure. It gives a reason to compare clustering with and without SASA, rather than assuming in advance that the two variables are interchangeable.

PCA provides a second description of the feature relationships. After standardization, the first component accounts for 51.8% of total variance and the second for 36.3%. Their combined share is 88.1% (fig. 3). Much of the variation lies in two directions, but the spectrum does not show whether smaller distinctions affect cluster boundaries. The later feature comparison addresses that question directly. Per-replica distributions are retained in fig. 6; they describe the recorded samples without equilibrium weighting.

6.3 Dependence between saved observations

The apparent sample size changes substantially when temporal dependence is considered. The correlation estimator returns provisional ESS values between 3.2 and 28.7 for the 12 replica–descriptor series, each containing 100 001 saved frames. All 12 estimates fail at least one of the specified reliability

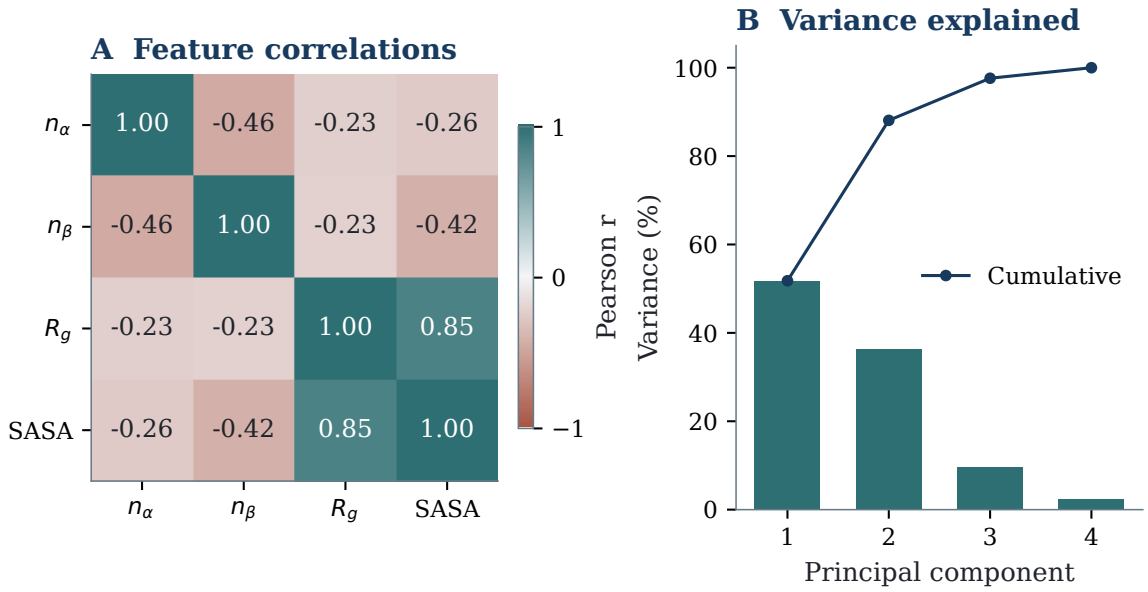


Figure 3: Relationships between the four descriptors in the pooled, unweighted Aβ40 sample. Left: Pearson correlation matrix. Right: variance explained by standardized PCA, with cumulative variance shown by the line. These summaries describe the biased observations, not equilibrium probabilities.

checks. The estimates therefore signal a problem for independent-frame reasoning; they do not provide precise counts of independent conformations.

The ACF curves show persistent correlations and differences between replicas (fig. 4). The accompanying diagnostics also record shifts between the means of the first and second halves of some series. Assessing uncertainty as if each frame were independent would be poorly supported by these results. They do not, however, constitute a new assessment of convergence for the full published trajectories: the complete bias history and original convergence diagnostics are unavailable.

6.4 What changes when SASA is removed

All 50 fits met the configured numerical convergence criterion. Together they cover two feature sets, five component counts and five paired initialization seeds. The pilot took 7.01 s and reached a sampled process-tree peak of 660.2 MiB. The models nevertheless produce different assignments across component counts and initializations. Figure 5 and table 3 compares both sources of variation on the same frame identities.

Components	Median ablation ARI	Observed range
2	0.690	0.188–0.868
3	0.864	0.862–0.868
4	0.607	0.548–0.622
5	0.531	0.382–0.696
6	0.457	0.294–0.461

Table 3: Agreement between feature sets across the five paired initializations. Component counts were specified for comparison, not selected as physical numbers of states.

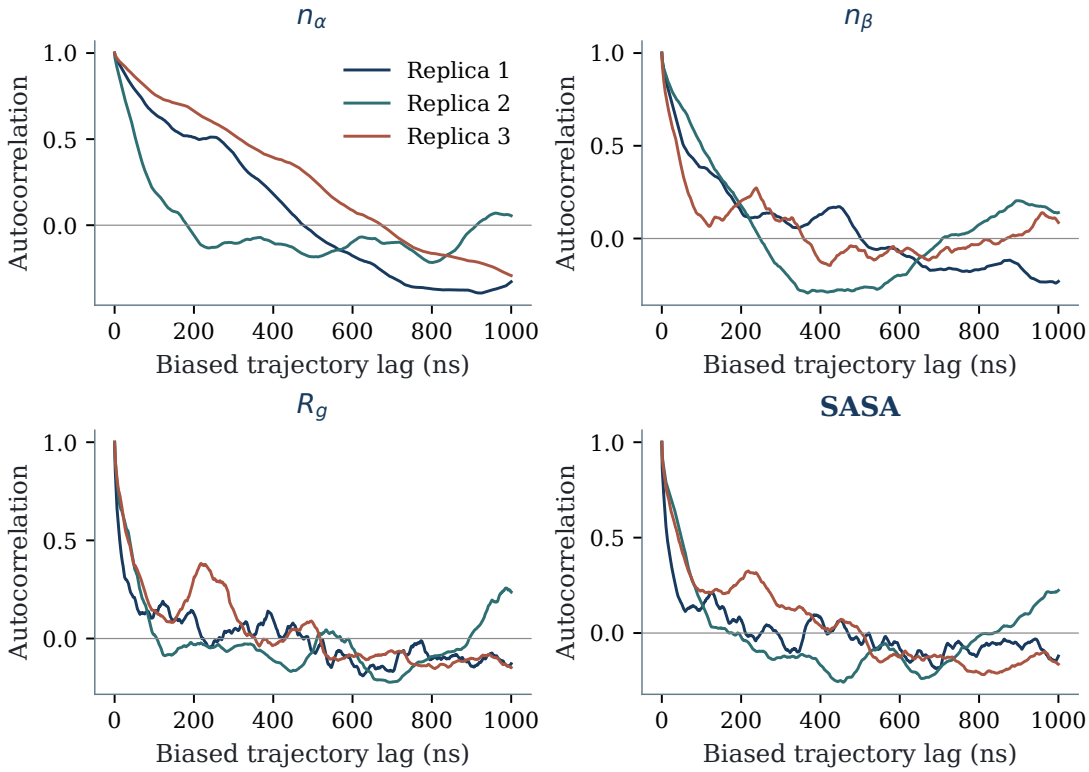


Figure 4: Autocorrelation functions calculated separately for each replica and descriptor. For readability the figure displays a subset of lag points over the first half of the available window; estimation uses the full ACF sequence. The horizontal axis follows biased trajectory time and is not an unbiased physical relaxation time.

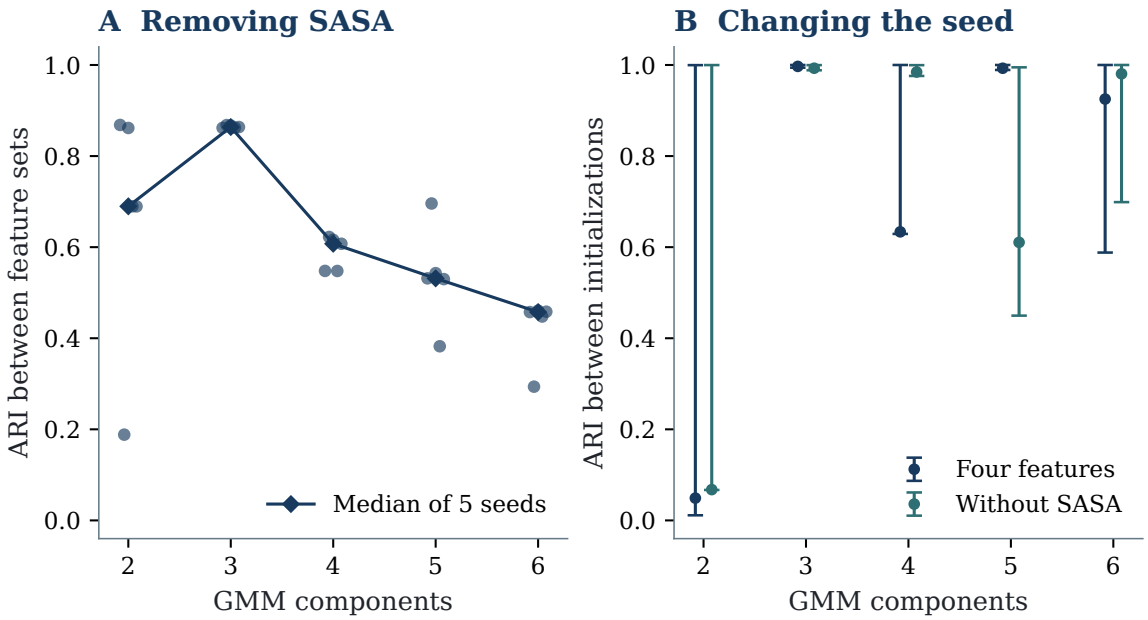


Figure 5: The independent GMM comparison on biased descriptor observations. Left: ARI between four-feature and SASA-removed assignments for five paired seeds; diamonds denote medians. Right: within-feature-space ARI for ten seed pairs. Bars span observed minima and maxima, not confidence intervals. Every model uses the same fitting subset, and evaluation includes fitting rows.

Changing the feature set

Paired fits use the same observations and seed. Their ARI measures how assignments change when SASA is excluded.

Changing the initialization

Fits within one feature space retain all descriptors in that representation. Their ARIs measure variation from initialization alone.

Reading the comparisons together helps distinguish a feature effect from variation already present before a descriptor is removed.

The three-component fits give cross-representation ARIs of 0.862–0.868. After maximum-overlap label matching, approximately 95.3% of frames retain corresponding labels, using the median over seeds. Within-space agreement is also high: 0.994–1.000 for four descriptors and 0.988–1.000 without SASA. At two components, the corresponding within-space ranges extend down to 0.011 and 0.067. A single fit would miss this variation; all seed comparisons are shown in fig. 7.

Removing SASA preserves much of the three-component partition in this fixed biased sample, while other counts show lower or more variable agreement. Equilibrium populations, weighted summaries and paper-DPC robustness remain uncomputed.

7 Discussion

7.1 A feature decision depends on the fitted model

The completed experiment began with an apparently simple reason to reduce the feature set: radius of gyration and SASA are strongly correlated. The comparison shows why that observation alone is insufficient. Correlation summarizes a relationship over the whole sample, whereas clustering assigns individual observations using the geometry of all retained features. The assignments can change when a feature is removed even if that feature shares much of its variation with another one.

The three-component mixtures retained high agreement in this pilot, both across feature sets and across the tested initializations. The other component counts did not give the same pattern. This does not select three as the physical number of conformational states. It shows that the answer to the feature-removal question depends on how the data are represented by the chosen model. Keeping all tested component counts makes this dependence visible instead of reporting only the most stable case.

The fixed fitting sample helps isolate the comparison but also limits it. Both feature sets use the same observations and the same scaling for shared features. Differences are therefore not caused by selecting a new sample for each fit. However, the experiment says nothing directly about how those results would change with another fitting subset. Initializations and sample selection are different sources of uncertainty, and only the former was varied here.

7.2 Connecting the extension to the published workflow

The primary study uses density-peak discovery, consensus and reweighting to characterize conformational families [2]. The GMM pilot examines a narrower statistical question using the available descriptors. Its agreement scores neither reproduce the paper's family populations nor test the reported difference between Aβ40 and Aβ42. They also give no direct information about aggregation rates or toxicity.

A common difficulty links the two approaches: a numerical family still needs a physical interpretation. A stable Gaussian partition might summarize a broad descriptor distribution rather than separate metastable states. The consensus baseline likewise compares marginal descriptor distributions and can miss differences in their joint dependence. Structures, contacts and alternative representations would help examine these possibilities. An independently constrained ensemble could provide another comparison, provided the structures, weights and observable definitions are compatible [6].

7.3 Sampling, weighting and the next comparison

The temporal diagnostics limit the uncertainty claims available from the present data. Randomly drawing fitting frames does not remove correlation from the process that generated them. All replica–descriptor series fail at least one estimator check, so the small provisional ESS values should be treated as warnings rather than calibrated effective counts. Repeating fits across fitting subsets and time blocks would address sampling sensitivity more directly [11].

Weighting introduces a further issue. Grid completeness establishes that the FES can be queried, but not that its scale or construction is appropriate for the chosen estimator. Once those details are verified, unequal weights may concentrate the estimate on relatively few observations. That effect must be assessed separately from seed variation and temporal dependence. With only three replicas, uncertainty based on differences between replicas will also need cautious interpretation. Saved labels,

tables and numerical tests make the completed calculations checkable while this provenance remains unresolved [4, 30].

8 Internship outcomes and skills

8.1 Learning the molecular context

The internship required more than applying a familiar statistical method to a new table. The assigned programme began with the concepts needed to interpret the data: amino acids, peptide bonds, protein structure and intrinsic disorder. Force fields, molecular dynamics and dihedral angles connected those concepts to simulation. The final task brought them together through examination of a clustering analysis and an independent investigation.

I used GROMACS, VMD and Python during the internship. Learning to use GROMACS was one of the more challenging parts of the internship for me. I found the process demanding at first, but I also enjoyed becoming more familiar with the software.

Saved configurations need not be independent observations, and their frequency in a biased simulation differs from their equilibrium probability. These points connect the molecular setting to statistics while showing why the origin of the data matters.

8.2 Turning an observation into an analysis

The Python project provides a concrete example. The descriptor analysis found a strong relationship between radius of gyration and SASA. Rather than treating this as a reason to remove SASA immediately, the independent experiment compared fitted assignments with and without it. The same fitting rows kept the comparison focused on the feature choice. Repeated initializations exposed variation that a single fit would have hidden.

Small programming decisions also became explicit parts of the analysis. Feature scaling is fitted on a specified sample and reused for later observations. A test checks that a large evaluation value cannot alter that transformation. Original frame identities are retained so that label arrays refer to the same observations. These details determine whether a difference comes from the method under investigation or an unintended change in preprocessing.

Fit once, reuse the same scale — RobustCluster-Aβ / Python

```python
def fit_scaler(fit_values, values):
    fit_values, values = finite(fit_values), finite(values)
    mean, scale = fit_values.mean(axis=0), fit_values.std(axis=0)
    if np.any(scale <= 0):
        raise InvalidInput("Constant fit feature")
    return (values - mean) / scale, {"mean": mean.tolist(), "scale": scale.tolist()}
```

The modules, tests, configurations, source tables and plots document the work produced. Successful execution remains distinct from establishing the physical meaning of a cluster.

8.3 My molecular-software interface projects

I also created two personal software projects, VMD MCP and GROMACS MCP. They use the Model Context Protocol to expose molecular-software operations through structured inputs and outputs. They are distinct from the Aβ40 descriptor analysis, but provide further examples of connecting scientific programs with Python interfaces.

VMD MCP: frame measurements

The inspected source constructs Tcl commands for measurement and rendering. Its gyration function passes an atom selection and mass-weighting option to VMD, then returns frame-indexed values.

GROMACS MCP: execution interface

The inspected source includes preparation and analysis helpers. Its background-run interface checks paths and records output in a log.

Contributor histories remain in the repositories. Appendix G links to the source versions inspected and pairs the code excerpts with genuine screenshots.

These interfaces organize requests; source code alone does not establish that a calculation ran. No reported Aβ40 trajectory is attributed to them. Comparing their outputs with the supplied descriptors would also require compatible selections and units: the inspected VMD gyration function documents angstroms, whereas the table is labelled in nanometres.

8.4 What can be carried forward

The report brings together the assigned learning programme, practical software use and calculations with saved outputs. Distinguishing them gives a clearer account than treating every planned task as completed. GROMACS and VMD use belongs to the practical experience; the descriptor diagnostics and GMM comparison belong to the recorded analysis.

The reusable outcome is a project whose steps can be examined. Each empirical figure has a source table, each calculation has settings, and each unavailable result has a reason. This organization can support further work once the FES units and construction are established. A Leonardo submission template is included for preparation, but no Leonardo job records exist. The reported computations remain local; further simulations or structural analyses will need their own execution records and validation.

9 Conclusions and future work

The internship materials support a study of how analysis choices affect the grouping of Aβ40 descriptors. Checking the supplied script against the published method identified problems in parsing, residual-cluster handling, FES interpolation, replica pooling and weighted summaries. The modular implementation makes the revised operations testable while preserving the original sources.

The independent GMM experiment gives the main completed comparison. Removing SASA largely preserves the tested three-component partition, but agreement changes at other component counts and with initialization. The Rg–SASA correlation therefore cannot decide the feature choice by itself. The experiment concerns one fitting subset and the biased descriptor sample; broader conclusions require further sampling comparisons and valid equilibrium weighting.

The next reproduction step is to establish the FES energy units and how the replicas were combined. A bounded real-data smoke run can then test the pipeline and measure resources before larger runs retain the published parameters. Seed, feature and Wasserstein-scaling experiments should compare the same frame identities so that changes in assignments have a clear meaning.

Additional inputs would enable distinct extensions. Verified bias-offset data would support time-dependent reweighting. Atomic trajectories with a reliable frame map would allow representative structures and coordinate-based analyses using tools such as MDAnalysis and ShapeGMM [29, 31]. An external ensemble comparison would require compatible observable definitions and appropriate data access.

The report's contribution is a documented analysis that can be inspected, with a completed test of feature sensitivity and explicit limits on what remains unknown. Its code and saved outputs provide a starting point for the remaining calculations. Equilibrium conformational families and their biological interpretation must follow from those calculations, rather than from the appearance of a stable partition alone.

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A Full parameter table

Table 4: Reproduction settings and execution profiles. These parameters are implemented but do not imply a completed empirical run.

Parameter	Value	Evidence / role
Replicas	3	Primary paper; observed table segmentation
Analysis interval	1–3 μs	Published baseline; inclusive source-labelled ps boundaries
Temperature	310 K	Published weighting parameter; no new MD run
Bias factor γ	10	Published weighting parameter
Molar gas constant	0.008314462618 kJ mol^{-1} K^{-1}	Implementation unit convention
CV conversions	$\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$	Published descriptor scaling
Baseline features	n_α, n_β, R_g, SASA	Pooled min–max normalization
Initial d_c	1.3	Published baseline
Cutoff multiplier	0.9	Published 10% reduction
Center-count scan	10 down to 2	Published baseline
Minimum cluster size	300 frames	Published acceptance criterion
Minimum mean silhouette	0.3	Published acceptance criterion
Failure stopping count	5	Consecutive iterations with no accepted cluster
Distance storage	float32	Source-informed; blockwise construction
Consensus	Average linkage	Marginal Wasserstein norm
Elbow smoothing	$s = 5$	Supplied source heuristic
Elbow search range	First 90% of merges	Supplied source heuristic
Final assignment	One nearest-centroid pass	No iterative refitting
Master seed	42	Default project configuration; recorded per run
Smoke / medium groups	3,000 / 10,000	Reduced execution profiles
Smoke / medium discovery groups	2 / 3	Not full-production evidence
Full group size	About 30,000	All groups; full execution disabled
Seed counts	2 smoke; 5 pilot/medium; 20 full default	Full robustness configurable; not performed
Bootstrap draws	500	Conditional moving-block extension; not performed on real weights
Resource policy	Bounded sequential local workers	Full execution held for review

Completed descriptor-only GMM pilot

The following settings belong to the successful independent experiment, not to the published DPC baseline. The fit sample and initializations were recorded before execution.

Table 5: Settings of the completed GMM experiment, as recorded in its run manifest.

Parameter	Value	Evidence / role
Fit sample	1,000 per replica	3,000 rows; shared across models
Sampling seed	20261007	Local NumPy generator
Feature spaces	Four descriptors; omit SASA	Same scaling for shared features
Scaling	Fit-sample mean and SD	No evaluation-data refit
Components	2, 3, 4, 5, 6	All counts retained in comparison
Initialization seeds	42, 43, 44, 45, 46	Paired across representations
Initializations per fit	1	50 total fitted models
Covariance	Full	Regularization 10^{-6}
Convergence settings	500 iterations; tolerance 10^{-3}	All fits converged numerically
Evaluation	300,003 rows	Includes fit rows; descriptive comparison
Resource limits	600 s; 4 GiB	Sequential local worker

B Mathematical derivations and estimator checks

Why the sign of the final-bias correction is negative in FES form

Under a static bias V, the observed configuration density is proportional to $\exp[-\beta_T(U+V)]$. Multiplying by $\exp(\beta_T V)$ recovers relative canonical density. The WT relation $V \simeq -(1-1/\gamma)F + C$ therefore gives the negative free-energy sign in eq. (2.2). Applying $\exp(-\beta_T V)$ would reverse the correction. This derivation assumes that using a final bias is appropriate for the analyzed interval; it is not a proof that the supplied trajectories satisfy that assumption.

Equal-replica mixture and cluster moments

Let $\widetilde{w}_{ri}$ sum to one within replica r. The pooled expectation of A is $(1/3)\sum_r \sum_i \widetilde{w}_{ri} A_{ri}$. Cluster relevance uses A equal to a membership indicator. For a conditional cluster mean, divide the pooled weighted sum of the descriptor by the pooled membership mass. Computing three conditional means and averaging them equally instead would answer a different question when cluster populations differ across replicas.

Bilinear interpolation test

For a cell with fractional coordinates $u, v \in [0,1]$, interpolation combines its four corner values with weights $(1-u)(1-v)$, $u(1-v)$, $(1-u)v$, and uv. Their sum is one. Any surface $a + b\alpha + c\beta + d\alpha\beta$ is reproduced exactly within floating-point tolerance. The test suite includes such a mixed-term surface, reordered input rows, boundaries and invalid-grid cases, providing stronger evidence than checking a single grid node.

C Additional diagnostics and interpretation

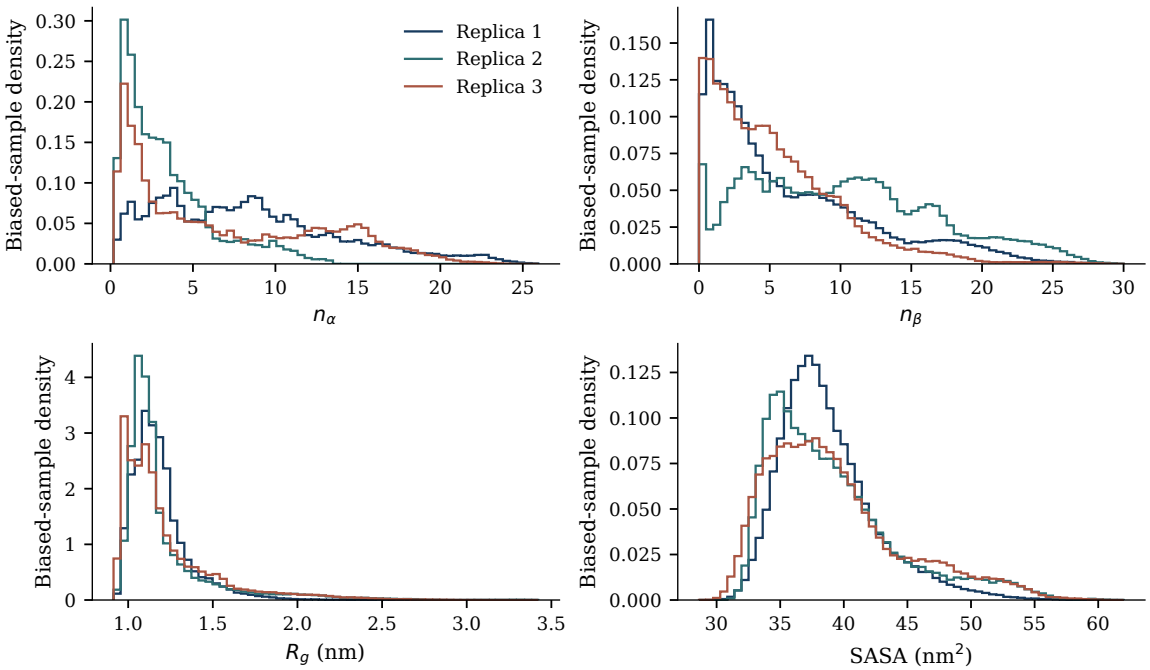


Figure 6: Per-replica distributions of the recorded descriptors. Each descriptor uses common bin edges, with separate density normalization for each replica. The curves retain the simulation bias and are not equilibrium distributions. Units follow the source labels.

Replica	Descriptor	g	τ_{int} (ns)	$N_{\mathrm{eff,corr}}$	Half shift / SD
1	n_α	22067	220.7	4.5	1.28
1	n_β	11079	110.8	9.0	0.83
1	R_g	3679	36.8	27.2	0.27
1	SASA	3490	34.9	28.7	0.36
2	n_α	6259	62.6	16.0	0.37
2	n_β	10843	108.4	9.2	0.41
2	R_g	3951	39.5	25.3	0.33
2	SASA	5325	53.2	18.8	0.17
3	n_α	31742	317.4	3.2	1.61
3	n_β	5880	58.8	17.0	0.71
3	R_g	6259	62.6	16.0	0.88
3	SASA	9958	99.6	10.0	0.87

Table 6: All recorded correlation estimates, each flagged as unreliable by at least one stated check. g denotes statistical inefficiency. The half-window shift is the absolute difference of means divided by the full-series standard deviation. The time axis belongs to the biased trajectory.

The saved tables contain per-replica raw descriptor counts, means, standard deviations, extrema, common-bin histograms, the full four-feature correlation matrix, PCA transformation parameters and explained variance, and the reported correlation-estimator diagnostics. The ACF figure is a

plotting subsample, not the series used to estimate g. No weights are present in these real-data diagnostic summaries.

The half-window mean-shift diagnostic is descriptive. A large shift can arise from slow fluctuations, nonstationarity, or changing occupation of sampled regions. Its threshold is an explicit project heuristic. A low shift alone would not establish convergence, and no threshold outcome can recover an unbiased time scale from WT-MetaD time. Future evaluation should preserve these distinctions when adding weighted estimates.

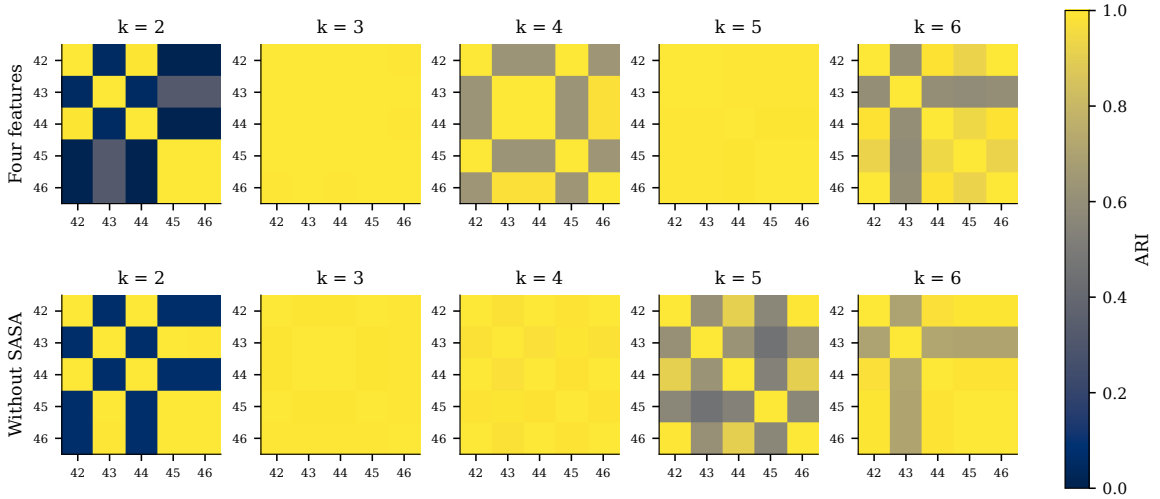


Figure 7: All pairwise seed ARIs from the GMM pilot. Columns show each tested component count; rows distinguish the four-feature and SASA-ablated representations. The diagonal is identity. All fits converged, so no entries are omitted. Similar numerical stopping status can coexist with different partitions.

D Code audit summary

The project inventory records immutable source hashes, table dimensions and missing inputs. The method map links each step to paper evidence, source behavior, the modular implementation and validation tests. The decision log records scientific category and expected effect. The open-questions register prevents uncertain provenance from being converted into hidden defaults.

Scientific V2 is absent, so the audit contains no invented intermediate diff. The supplied original remains excluded from execution and lint collection because its syntax error and top-level writes are part of the preserved evidence. New code is tested independently. Public source packaging excludes the raw table and original script until their redistribution rights are established.

Codex assisted with software refactoring, numerical checks, document preparation and layout. Reported numerical observations come from recorded Python runs; the illustrated cover and two conceptual editorial artworks were generated with the built-in image-generation tool, and the source-code screenshots are actual browser captures. This assistance is separate from the published methodology and the original supplied materials. The generation tool does not expose a verifiable model-version identifier; no specific Image 2.5 execution claim is made. Exact prompts and image hashes are retained in `docs/ARTWORK_PROMPTS.json` and `report/artwork_provenance.json`. The supplied inspiration image guided the print texture and restrained palette, not the scientific content. The cover typography is part of the generated image; its institutional and personal details were visually checked. The University of Messina logo is a separate image supplied by the student and placed without generative reconstruction.

E Reproducibility instructions

The project source, tests, configurations and this report are available at RobustCluster-Aβ on GitHub. The Python package and LaTeX sources can be inspected separately or downloaded together using GitHub's *Code / Download ZIP* option. The repository guide distinguishes code verification and report regeneration from analyses requiring restricted inputs.

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lock.txt
python -m pip install --no-deps -e .
pytest -q
ruff check src tests scripts
robustcluster validate --config config/paper_reproduction.yaml
robustcluster diagnostics --config config/robustness.yaml
```

The raw file must first be obtained and checked against its documented SHA256. The standalone commands and Snakemake workflow use the same package functions. The report imports successful diagnostics, FES-audit, GMM and figure manifests with matching hashes. Its generated editorial artwork and real browser screenshots have separate provenance records. LuaLaTeX, Biber and latexmk build the supplied report sources; current paths and tool versions are recorded in the build manifest.

For later production, establish the recovered FES energy units and construction provenance, run smoke validation, review resource measurements, and create a reviewed full-run configuration. HPC templates require an actual allocation and explicit settings. Do not copy account or partition values from unrelated examples, and preserve job evidence before attributing calculations to Leonardo [32].

F Software and environment table

Component	Version in diagnostic manifest
Python	3.12.6
numpy	2.5.3
scipy	1.18.1
scikit-learn	1.9.1
pandas	2.3.3
matplotlib	3.11.2
robustcluster-abeta	0.1.0

Table 7: Actual software versions for the reported local descriptor diagnostics; the GMM pilot manifest records the same versions. They do not describe the original authors' simulation environment. The complete dependency lock and report-tool versions are separate reproducibility artifacts.

The checked bibliography contains journal articles and official documentation records; its matrix links each citation to its question, method, relevance, and report section. Online-first dates and issue years are distinguished. References whose publisher verification remained unresolved are documented outside the BibTeX file rather than filled from memory.

G Companion source-code evidence

The following images are genuine browser captures from public, commit-specific GitHub source pages on 7 October 2026. They document implementation, not simulation execution. The full images are preserved unchanged; short accompanying searchable listings are a separate presentation format.

02 Measure each saved frame — VMD MCP / source lines 250–266

```
    weight = "weight mass" if mass_weighted else ""
    tcl = f"""
{_load_block(structure, trajectory)}
set sel [atomselect $mol "{_tcl_quote(selection)}"]
set n [molinfo $mol get numframes]
for {{set i 0}} {{$i < $n}} {{incr i}} {{
  $sel frame $i
  puts "{MARKER} frame=$i rgyr=[measure rgyr $sel {weight}]"
}}
quit
"""
    res = _run_tcl(tcl)
    series = [
        {"frame": int(m["frame"]), "rgyr": float(m["rgyr"])}
        for m in res.get("markers", [])
        if "rgyr" in m
    ]
```

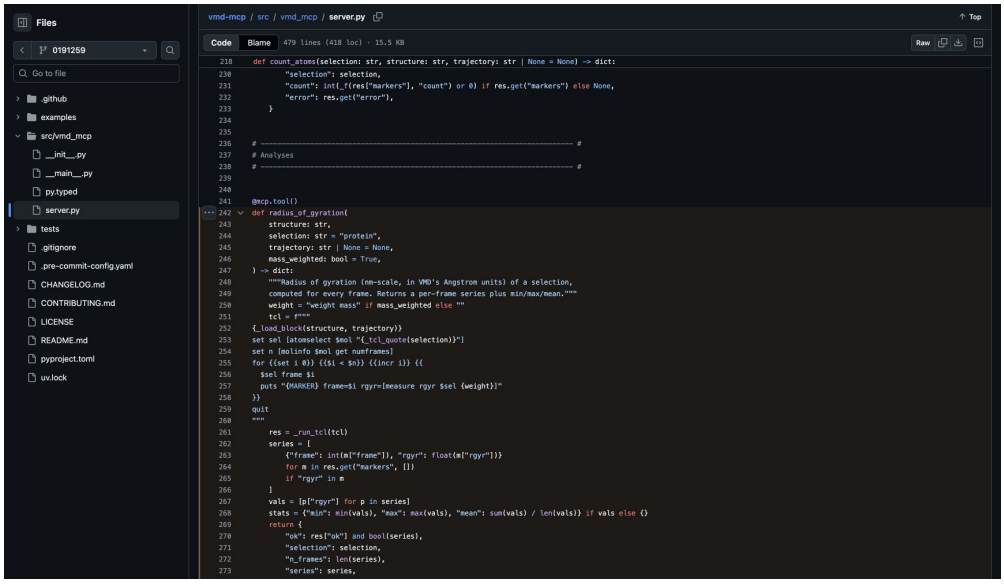


Figure 8: VMD MCP source at commit 0191259: the gyration wrapper constructs a Tcl frame loop and parses marked values. The screenshot shows the repository tree, source path and real line numbers. The function's documented unit convention must be checked before comparison with other tools. Open the inspected source.

03 Validate before starting a job GROMACS MCP / source lines 1452–1467

```
1452        wd, err = _workdir_or_error(workdir)
1453        if err:
1454            return err
1455        assert wd is not None
1456        if ntomp is not None and ntomp <= 0:
1457            return _error("ntomp must be positive when provided")
1458        if nsteps is not None and nsteps < 0:
1459            return _error("nsteps must be non-negative when provided")
1460        safe_tpr = _safe_file_arg(wd, tpr, "tpr", must_exist=True)
1461        safe_deffnm = _safe_deffnm(deffnm)
1462        args = [GMX_BIN, "mdrun", "-v", "-s", safe_tpr, "-deffnm", safe_deffnm]
1463        if ntomp is not None:
1464            args += ["-ntomp", str(ntomp)]
1465        if nsteps is not None:
1466            args += ["-nsteps", str(nsteps)]
1467        args += _validate_arg_tokens(extra_args, "extra_args")
```

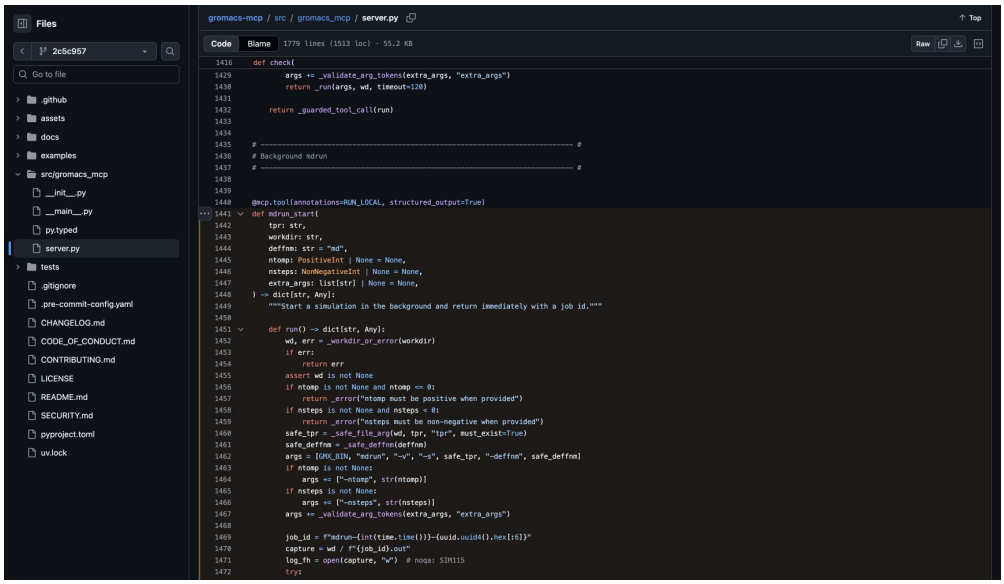


Figure 9: GROMACS MCP source at commit `2c5c957`: the background-run wrapper validates inputs, constructs the `mdrun` command and prepares an output log. This source view does not establish that an Aβ40 simulation was launched through the tool. Open the inspected source.